MATH 128 - HW3 - Jie Wen

Problem 1: Problems from textbook Section 1.3

3

a. Find n such that $|\pi - 4P_n(1)| < 10^{-3}$

The error bound after n terms is the magnitude of the next term: $|R_n| < \frac{1}{2n+1}$.
The error in approximating π is $|4P_n(1) - \pi| < \frac{4}{2n+1}$.

To ensure $\frac{4}{2n+1} < 10^{-3}$, solve $2n + 1 > 4000$, so $n > 1999.5$. The smallest integer n is 2000.

b. Find n such that $|\pi - 4P_n(1)| < 10^{-10}$

Similarly, to ensure $\frac{4}{2n+1} < 10^{-10}$, solve $2n + 1 > 4 \times 10^{10}$, so $n > 2 \times 10^{10} - 0.5$. The smallest integer n is 2×10^{10}.

6

a. $\sin\left(\frac{1}{n}\right)$

Using the Taylor expansion:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Substitute $x = \frac{1}{n}$:

$$\sin\left(\frac{1}{n}\right) = \frac{1}{n} - \frac{1}{6n^3} + \cdots \sim \boxed{\frac{1}{n}}$$

b. $\sin\left(\frac{1}{n^2}\right)$

Substitute $x = \frac{1}{n^2}$:

$$\sin\left(\frac{1}{n^2}\right) = \frac{1}{n^2} - \frac{1}{6n^6} + \cdots \sim \boxed{\frac{1}{n^2}}$$

c. $\left(\sin\left(\frac{1}{n}\right)\right)^2$

Using the expansion from part (a):

$$\left(\sin\left(\frac{1}{n}\right)\right)^2 = \left(\frac{1}{n} - \cdots\right)^2 = \frac{1}{n^2} + \cdots \sim \boxed{\frac{1}{n^2}}$$

d. $\ln(n+1) - \ln(n)$

Using logarithmic properties:

$$\ln(n+1) - \ln(n) = \ln\left(1 + \frac{1}{n}\right)$$

Using the expansion:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

Substitute $x = \frac{1}{n}$:

$$\ln\left(1 + \frac{1}{n}\right) = \frac{1}{n} - \frac{1}{2n^2} + \cdots \sim \boxed{\frac{1}{n}}$$

7

a. $\lim_{h\to 0} \frac{\sin h}{h} = 1$

Using the Taylor expansion:

$$\sin h = h - \frac{h^3}{6} + \frac{h^5}{120} - \cdots$$

Then,

$$\frac{\sin h}{h} = 1 - \frac{h^2}{6} + \frac{h^4}{120} - \cdots$$

So,

$$\frac{\sin h}{h} - 1 = -\frac{h^2}{6} + O(h^4) \sim \boxed{O(h^2)}$$

b. $\lim_{h\to 0} \frac{1-\cos h}{h} = 0$

Using the Taylor expansion:

$$\cos h = 1 - \frac{h^2}{2} + \frac{h^4}{24} - \cdots$$

Then,

$$1 - \cos h = \frac{h^2}{2} - \frac{h^4}{24} + \cdots$$
$$\frac{1-\cos h}{h} = \frac{h}{2} - \frac{h^3}{24} + \cdots$$

So,

$$\frac{1-\cos h}{h} - 0 = \frac{h}{2} + O(h^3) \sim \boxed{O(h)}$$

c. $\lim_{h\to 0} \frac{\sin h - h\cos h}{h} = 0$

Using Taylor expansions:

$$\sin h = h - \frac{h^3}{6} + \frac{h^5}{120} - \cdots$$
$$\cos h = 1 - \frac{h^2}{2} + \frac{h^4}{24} - \cdots$$

Then,

$$\sin h - h\cos h = \left(h - \frac{h^3}{6}\right) - \left(h - \frac{h^3}{2}\right) + \cdots = \frac{h^3}{3} + O(h^5)$$
$$\frac{\sin h - h\cos h}{h} = \frac{h^2}{3} + O(h^4)$$

So,

$$\frac{\sin h - h\cos h}{h} - 0 = \frac{h^2}{3} + O(h^4) \sim \boxed{O(h^2)}$$

d. $\lim_{h\to 0} \frac{1-e^h}{h} = -1$

Using the Taylor expansion:

$$e^h = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \cdots$$

Then,

$$1 - e^h = -h - \frac{h^2}{2} - \frac{h^3}{6} - \cdots$$

$$\frac{1 - e^h}{h} = -1 - \frac{h}{2} - \frac{h^2}{6} - \cdots$$

So,

$$\frac{1 - e^h}{h} - (-1) = -\frac{h}{2} + O(h^2) \sim \boxed{O(h)}$$

8

a. Given that $\alpha_n = \alpha + O(n^{-p})$, there exists a constant $C > 0$ such that for all sufficiently large n,

$$|\alpha_n - \alpha| \leq Cn^{-p}.$$

Since $0 < q < p$, we have $n^{-p} \leq n^{-q}$ for all $n \geq 1$. Therefore,

$$|\alpha_n - \alpha| \leq Cn^{-p} \leq Cn^{-q},$$

which implies that $\alpha_n = \alpha + O(n^{-q})$.

b. The table below lists the values of $\frac{1}{n}$, $\frac{1}{n^2}$, $\frac{1}{n^3}$, and $\frac{1}{n^4}$ for the specified n:

n	$\frac{1}{n}$	$\frac{1}{n^2}$	$\frac{1}{n^3}$	$\frac{1}{n^4}$
5	0.2	0.04	0.008	0.0016
10	0.1	0.01	0.001	0.0001
100	0.01	0.0001	0.000001	0.00000001
1000	0.001	0.000001	0.000000001	0.000000000001

As n becomes large, all sequences converge to 0, but at different rates. The sequence $\frac{1}{n^4}$ converges fastest (decreases most rapidly with increasing n), followed by $\frac{1}{n^3}$, $\frac{1}{n^2}$, and $\frac{1}{n}$ converges slowest. This is because higher exponents in the denominator result in smaller values more quickly; for example, by $n = 1000$, $\frac{1}{n^4}$ is already on the order of 10^{-12}, while $\frac{1}{n}$ is only 10^{-3}. In big-O notation, these have rates $O(\frac{1}{n})$, $O(\frac{1}{n^2})$, $O(\frac{1}{n^3})$, and $O(\frac{1}{n^4})$, with higher powers indicating faster convergence to 0.

10

a. Given that $F_1(x) = L_1 + O(x^\alpha)$ and $F_2(x) = L_2 + O(x^\beta)$ as $x \to 0$, there exist constants $M_1, M_2 > 0$ and $\delta_1, \delta_2 > 0$ such that for $|x| < \delta_1$, $|F_1(x) - L_1| \leq M_1|x|^\alpha$, and for $|x| < \delta_2$, $|F_2(x) - L_2| \leq M_2|x|^\beta$.

Then,

$$F(x) = c_1F_1(x) + c_2F_2(x) = c_1L_1 + c_2L_2 + c_1(F_1(x) - L_1) + c_2(F_2(x) - L_2).$$

So,

$$|F(x) - (c_1L_1 + c_2L_2)| \leq |c_1||F_1(x) - L_1| + |c_2||F_2(x) - L_2|.$$

Assume WLOG that $\alpha \leq \beta$, so $\gamma = \alpha$. For $|x| < \min(\delta_1, \delta_2, 1)$,

$$|F(x) - (c_1L_1 + c_2L_2)| \leq |c_1|M_1|x|^\alpha + |c_2|M_2|x|^\beta \leq |c_1|M_1|x|^\alpha + |c_2|M_2|x|^\alpha = (|c_1|M_1 + |c_2|M_2)|x|^\alpha,$$

since $|x|^\beta \leq |x|^\alpha$ when $|x| < 1$. Thus, $F(x) = c_1L_1 + c_2L_2 + O(x^\gamma)$. The case $\beta < \alpha$ follows similarly.

b. Now,

$$G(x) = F_1(c_1x) + F_2(c_2x) = L_1 + O((c_1x)^\alpha) + L_2 + O((c_2x)^\beta) = L_1 + L_2 + O(|c_1|^\alpha|x|^\alpha) + O(|c_2|^\beta|x|^\beta).$$

Since $|c_1|^\alpha$ and $|c_2|^\beta$ are constants, this is $L_1 + L_2 + O(|x|^\alpha) + O(|x|^\beta)$. As in part **a.**, the sum is $O(x^\gamma)$, so $G(x) = L_1 + L_2 + O(x^\gamma)$.

11

From the Fibonacci recurrence, divide both sides by F_{n+1}:

$$x_{n+1} = 1 + \frac{1}{x_n}$$

Assume:

$$\lim_{n\to\infty} x_n = x$$

Taking the limit on both sides of the recurrence:

$$x = 1 + \frac{1}{x}$$

Rewriting:

$$x^2 - x - 1 = 0$$

Using the quadratic formula:

$$x = \frac{1 \pm \sqrt{1^2 + 4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

Since $x_n > 0$ for all n, take the positive root:

$$x = \frac{1 + \sqrt{5}}{2}$$

Problem 2: Problems from textbook Section 2.1

1

Evaluate at the endpoints:

- $f(0) = \sqrt{0} - \cos 0 = 0 - 1 = -1$
- $f(1) = \sqrt{1} - \cos 1 \approx 1 - 0.5403 = 0.4597$

Since $f(0) < 0$ and $f(1) > 0$, by the Intermediate Value Theorem, a root exists in ([0, 1]).

Iteration 1:

- $a_1 = 0, b_1 = 1$
- Midpoint: $p_1 = \frac{0+1}{2} = 0.5$
- $f(0.5) = \sqrt{0.5} - \cos(0.5) \approx 0.7071 - 0.8776 = -0.1705$

Since $f(0.5) < 0$ and $f(1) > 0$, the root lies in ([0.5, 1]).

Iteration 2:

- $a_2 = 0.5, b_2 = 1$
- Midpoint: $p_2 = \frac{0.5+1}{2} = 0.75$
- $f(0.75) = \sqrt{0.75} - \cos(0.75) \approx 0.8660 - 0.7317 = 0.1343$

Since $f(0.5) < 0$ and $f(0.75) > 0$, the root lies in ([0.5, 0.75]).

Iteration 3:

- $a_3 = 0.5, b_3 = 0.75$

- Midpoint: $p_3 = \frac{0.5+0.75}{2} = \boxed{0.625}$

4

MATLAB Code:

```
function p = bisection(f, a, b, tol)
    % Bisection method to find a root of f in [a, b] within tolerance tol.
    % Assumes f(a) and f(b) have opposite signs.
    while (b - a) > tol
        p = (a + b) / 2;
        if f(p) == 0
            return;
        elseif f(a) * f(p) < 0
            b = p;
        else
            a = p;
        end
    end
    p = (a + b) / 2;
end

% Define the function
f = @(x) x.^4 - 2*x.^3 - 4*x.^2 + 4*x + 4;

% Tolerance for accuracy (interval length < 0.02 ensures error < 0.01)
tol = 0.02;

% Solve for each interval
a1 = -2; b1 = -1;
root_a = bisection(f, a1, b1, tol);
fprintf('Root in [-2, -1]: %.4f\n', root_a);

a2 = 0; b2 = 2;
root_b = bisection(f, a2, b2, tol);
fprintf('Root in [0, 2]: %.4f\n', root_b);

a3 = 2; b3 = 3;
root_c = bisection(f, a3, b3, tol);
fprintf('Root in [2, 3]: %.4f\n', root_c);

a4 = -1; b4 = 0;
root_d = bisection(f, a4, b4, tol);
fprintf('Root in [-1, 0]: %.4f\n', root_d);
```

Output:

```
Root in [-2, -1]: -1.4141
Root in [0, 2]: 1.4141
Root in [2, 3]: 2.7266
Root in [-1, 0]: -0.7266
```

12

MATLAB Code:

```
% Define the function
f = @(x) (x+2) .* (x+1).^2 .* x .* (x-1).^3 .* (x-2);

% Known roots from the factored form
known_roots = [-2, -1, 0, 1, 2];

% Intervals to test
intervals = [-1.5, 2.5;  % a
             -0.5, 2.4;  % b
             -0.5, 3.0;  % c
```

```
            -3.0, -0.5]; % d

% Bisection method parameters
tol = 1e-10; % Tolerance
max_iter = 100; % Maximum iterations

fprintf('Bisection Method Convergence Analysis for f(x) = (x+2)(x+1)^2 x (x-1)^3 (x-2)\n\n');

for i = 1:size(intervals, 1)
    a = intervals(i, 1);
    b = intervals(i, 2);
    fa = f(a);
    fb = f(b);

    fprintf('Interval [%g, %g]:\n', a, b);

    % Check if the function has opposite signs at the endpoints
    if sign(fa) == sign(fb)
        fprintf('   Warning: f(a) and f(b) have the same sign. Bisection method may not converge to a root.\n');
        % The method might not converge or may converge to a root if f(a) or f(b) is zero.
        % We proceed with checks for zero endpoints.
        root_found = NaN;
        if fa == 0
            root_found = a;
            fprintf('   f(a) is exactly zero. Converges to x = %g (root at a).\n', root_found);
        elseif fb == 0
            root_found = b;
            fprintf('   f(b) is exactly zero. Converges to x = %g (root at b).\n', root_found);
        else
            fprintf('   Cannot apply standard bisection method on this interval.\n');
        end
    else
        % Apply the Bisection Method
        iter = 0;
        while (b - a) / 2 > tol && iter < max_iter
            c = (a + b) / 2; % Midpoint
            fc = f(c);

            if fc == 0
                % Root found exactly
                break;
            elseif sign(fc) == sign(fa)
                a = c;
                fa = fc;
            else
                b = c;
                fb = fc;
            end
            iter = iter + 1;
        end
        root_found = (a + b) / 2; % Final approximation
        fprintf('   Converges to x = %g after %d iterations.\n', root_found, iter);
    end

    % Identify which known root it converged to
    if ~isnan(root_found)
        [minDiff, idx] = min(abs(known_roots - root_found));
        fprintf('   This is the root at x = %g.\n', known_roots(idx));
    end
    fprintf('\n');
end
```

Output:

```
Bisection Method Convergence Analysis for f(x) = (x+2)(x+1)^2 x (x-1)^3 (x-2)

Interval [-1.5, 2.5]:
   Converges to x = 0 after 2 iterations.
   This is the root at x = 0.

Interval [-0.5, 2.4]:
   Converges to x = 7.85803e-11 after 34 iterations.
   This is the root at x = 0.

Interval [-0.5, 3]:
```

```
   Converges to x = 2 after 35 iterations.
   This is the root at x = 2.

Interval [-3, -0.5]:
   Converges to x = -2 after 34 iterations.
   This is the root at x = -2.
```

13

MATLAB Code:

```
% Bisection method for finding cube root of 25
f = @(x) x.^3 - 25;
a = 2;
b = 3;
tol = 1e-4;
max_iter = 100;

fa = f(a);
fb = f(b);

if fa * fb > 0
    error('The function does not change sign on the interval [a, b].');
end

iter = 0;
while (b - a) / 2 > tol
    iter = iter + 1;
    c = (a + b) / 2;
    fc = f(c);
    if fc == 0
        break;
    elseif fa * fc < 0
        b = c;
        fb = fc;
    else
        a = c;
        fa = fc;
    end
    if iter >= max_iter
        error('Maximum iterations reached without convergence.');
    end
end

root = (a + b) / 2;
fprintf('Approximation to cube root of 25: %.8f\n', root);
fprintf('Number of iterations: %d\n', iter);
```

Output:

```
Approximation to cube root of 25: 2.92401123
Number of iterations: 13
```